Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058,
India (Autonomous College Affiliated to University of Mumbai)

Mid Semester Examination
October 2025

Max. Marks: 30 | **Duration: 1Hr.**
Class: T. Y. COMP, CSE | **Semester: V**
Course Code: CE301 , CS301
Subject: Distributed Computing | **Date:06/10/2025**

Instruction:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary
(4) Mention the question number clearly while writing the answer
(5) Write answers to new questions on new page.

Course Outcomes: At the end of the course student will be able to:
1. Understand the fundamental concepts and design principles of distributed systems.
2. Analyze and apply synchronization techniques and algorithms in distributed environments.
3. Evaluate consistency models and replication strategies for distributed data management.
4. Design and assess distributed file systems and understand their real-world applications.
5. Explore emerging trends and technologies in distributed computing.

Q No.		Max. Marks	CO BL
Q1.	A smart healthcare system connects hospitals, pharmacies, ambulances, and diagnostic centers using a distributed architecture. The government must decide between: 1. A Centralized Client-Server system 2. A Peer-to-Peer system 3. A Middleware-based distributed system Evaluate each model for: 1. Reliability during emergencies (e.g., ambulance needing patient history) 2. Security of sensitive medical records 3. Performance during simultaneous access by multiple hospitals Use the healthcare analogy in your evaluation (e.g., patient records as shared resources, hospitals as nodes, middleware as coordinators). Based on your evaluation, argue which model provides the best balance for such a healthcare system. Justify with the help of Architecture.	10	1-5

Q2	A global airline system allows passengers, travel agents, and airports to communicate for booking tickets, seat allocation, and flight updates. Each request/response can be seen as a message. Design a distributed communication framework for this system by: 1. Selecting whether to use Message Passing, RPC, or RMI for core operations (e.g., seat booking, flight status) and justify your choice. 2. Proposing how message persistence and synchronicity should be managed (e.g., handling delayed confirmations, ensuring bookings are not lost). 3. Creating a mechanism for buffering and failure handling (e.g., if the central booking server crashes mid-transaction). Present your design using the airline analogy (messages as tickets, servers as airports, middleware as control towers) and explain how your solution ensures reliability and consistency. Write your answer with diagrams and justify with respect to scenario.	10	1-6
Q3	A smart hotel system allows guests to: 1. Order room service via a mobile app (RPC) 2. Check and update loyalty points using a remote banking service (RMI) Given Data:For RPC (Room Service Ordering): • Preparation time per dish = 10 minutes • Network latency per RPC call = 2 minutes (round trip) • Marshaling/unmarshaling overhead = 1 minute per call • Guest orders 3 dishes For RMI (Loyalty Points Service): • Time to execute checkPoints() = 2 seconds • Time to execute updatePoints() = 5 seconds • RMI network overhead = 1 second per call • Session initialization = 2 seconds • Guest performs 2 balance checks and 1 update Part A – RPC (5 marks) 1. Calculate the total time to receive all 3 dishes if RPC calls are sent sequentially. 2. Calculate the total time if RPC calls are sent in parallel. Part B – RMI (5 marks) 1. Calculate the total time for all RMI operations sequentially. 2. Suggest a way to optimize RMI calls to reduce total time and calculate the improved time.	10	1-3